A Joint Saliency Temporal–Spatial–Spectral Information Network for Hyperspectral Image Change Detection

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***Abstract*— Hyperspectral image change detection (HSI-CD) is a fundamental task in the field of remote sensing (RS) observation, which utilizes the rich spectral and spatial information in bitemporal HSIs to detect subtle changes on the Earth's surface. However, modern deep learning (DL)-based HSI-CD methods mostly rely on patch-based methods, which leads to spectral band redundancy and spatial information noise in limited receiving domains, thus ignoring the extraction and utilization of saliency information and limiting the improvement of CD performance. To address these issues, this article proposes *a joint saliency temporal–spatial–spectral information network* (STSS-Net) for HSI-CD. The principal contributions of this article can be summarized: 1) we have designed a spatial saliency information extraction (SSIE) module for denoising based on distance from center pixels and spectral similarity of the substance, which increases the attention to spatial differences between similar spectral substances and different spectral substances; 2) we have designed a compact high-level spectral information tokenizer (CHLSIT) for spectral saliency information, where the high-level conceptual information of changes in spectral interest can be represented by nonlinear combinations of spectral bands, and redundancy can be removed by extracting high-level spectral conceptual features; and 3) utilizing the advantages of CNN and transformer architectures to combine temporal–spatial–spectral information. The experimental results on three real HSI-CD datasets show that STSS-Net can improve the accuracy of CD and has a certain improvement in the detection of edge information and complex information.**

***Index Terms*— Attention, change detection, convolutional neural networks (CNNs), hyperspectral image (HSI), saliency information, transformer.**

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I. Introduction

HYPERSPECTRAL image change detection (HSI-CD) is an important technology that utilizes bitemporal hyperspectral images (HSIs) of the same area to identify changes in natural scenes. Unlike many other remote sensing (RS) images, HSIs with hundreds of continuous spectral bands from ultraviolet to infrared have high spectral and spatial resolution. In recent years, with the increasing availability of RS satellite images, HSI-CD has found extensive applications in various fields, including but not limited to agricultural development, vegetation monitoring, environmental assessment, urban planning, and disaster evaluation, etc [1], [2], [3], [4].

Mainstream methods in the early development of CD involved algebra [5], transformation [6], and classification [7], [8]. These methods demanded extensive image preprocessing and relied on image pixels for calculations. Changes were obtained by differentiating dual temporal pixel pairs and applying appropriate thresholds. However, these methods only relied on handcrafted shallow features, which were inadequate in representing crucial information in HSIs. Consequently, the accuracy of the obtained CD results was often subpar. Nevertheless, the introduction of new methods has propelled the progress of CD.

With the rapid expansion of big data in recent years, the prominence of deep learning (DL) methods has soared. RS image CD has extensively adopted deep neural networks (DNNs) [9], convolutional neural networks (CNNs) [10], recurrent neural networks (RNNs) [11], transformer [12], and various other DL networks.

With the rapid development of spectral imaging technology, HSIs are increasingly being used for Earth observation. It offers the advantage of capturing more detailed ground object information. Consequently, bi-temporal HSI data have emerged as a reliable source for change detection. However,

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unlike ordinary CD tasks, HSI-CD technology is still in the early stages of development. The main challenges can be summarized as follows.

1) Limited receiving domain. For the CD task of HSIs, the shortage of labeled training data has always been a typical problem [13]. Therefore, for the extraction of spatial context information, current methods usually use patch-based methods, which focus more on local information in limited spatial information [14], [15]. However, the limited acceptance domain results in the loss of some global information. As shown in Fig. 1, the restricted domain gradually expands from left to right (white, gray, yellow boxes), which in turn causes excessive neighborhood information to interfere with the judgment of center pixel detection.
2) Hyperspectral band redundancy. HSIs have a wide range of bands, each with different contributions to CD. Bands that don't contribute much to change detection can impact accuracy, while excessive data can strain computer calculations [16], [17], [18]. As shown on the right side of Fig. 1, due to the increase in patch size and the presence of neighboring spectral information, there is a problem of increasing redundant spectral information. How to fully utilize the channel advantages of HSIs and remove redundancy is one of the challenges faced by HSI-CD.
3) Joint temporal–spectral–spatial information. For certain objects, it is difficult to use only spectral/spatial/temporal features for CD. We need to consider the correlation between these features [19], [20]. Recent research suggests that extracting temporal–spectral–spatial features by combining spectral information, spatial information, and temporal information can enhance performance [21]. At present, the main strategy for joint feature extraction is to sequentially extract temporal, spatial, and spectral information in series to achieve the purpose of fused features to directly extract spectral–spatial features using 3D-CNN [22] or to use attention mechanisms for feature fusion [23], [24]. So how to combine the advantages of CNN and transformers to combine temporal spectral spatial information is a worthwhile research direction.

Due to contextual modeling within spatial, spectral, and temporal ranges, extracting effective and saliency information is crucial for identifying changes in HSIs. To address the above challenges, Hong et al. [25] can learn spectral sequence information from adjacent bands of HSIs for hyperspectral classification tasks, but due to the lack of attention to spatial and spectral redundant noise, the effect fluctuates greatly, and effective information cannot be effectively utilized. Gong et al. [26] have proposed a spatial attention mechanism based on distance using Gaussian sampling to address the impact of neighborhood spatial noise. However, it ignores the importance of different background pixels in the neighborhood. Although it stabilizes the effect under different patches, it also loses some global information. To solve the problem of spectral band redundancy, a fast and slow changing band selection method was proposed, which directly selects

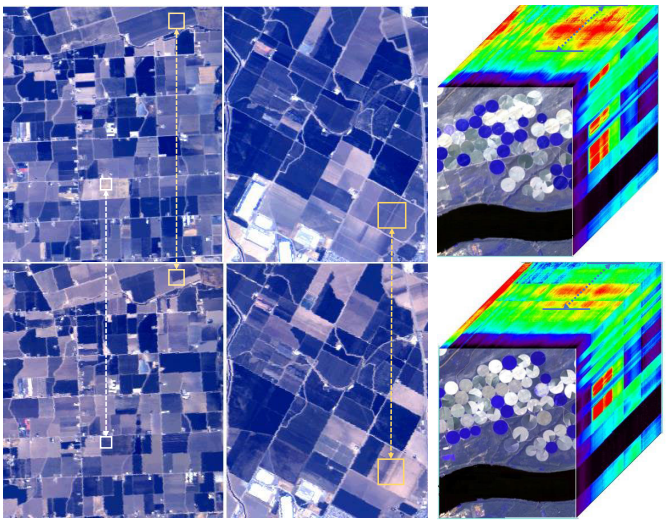


Fig. 1. Explanation on the necessity of extracting spatial and spectral significance information.

effective spectral band information for feature extraction [18]. This method is simple and effective but may overlook the situation of similar spectra of different substances, resulting in poor CD performance in complex situations. The main architecture of the method proposed in Article [23] is the transformer, which combines spatial information and spectral features to generate joint features through concatenation of transformer encoders (TEs) and reshaping. A new attention mechanism is designed to extract dual temporal features.

To address the above challenges, we denoise spatial contextual information based on spectral similarity and distance information. Then, utilizing the advantages of CNN in extracting local and detailed information [27]. The above part expands the range of local information and improves spatial saliency. Inspired by BIT-CD [28], the high-level conceptual information of changes in spectral interest can be represented by nonlinear combinations of several spectral bands. We first eliminate redundancy by compressing spectral information. Then, different levels of attention to high-level conceptual spectral bands are maintained through channel attention. Finally, matrix multiplication is used to utilize the spectral band information of neighboring pixels in the same band. Efficient modeling of spectral global sequence context in dual time HSIs during transformers.

Our job is not to model the dense sequence relationships between spectral bands in pixel spectra through simple reshape HSIs, but to express the input image as several high-level concept tokens and model the global sequence and temporal context of the spectrum based on high-level spectral concept tokens. In addition, we jointly represent temporal–spectral–spatial features by utilizing the relationship between the spatial features of each pixel and temporal–spectral tokens.

We can summarize our work in the following way.

1) We propose a joint saliency temporal–spatial–spectral information network (STSS-Net). This network combines temporal–spectral–spatial feature information to extract CD-related saliency information through three

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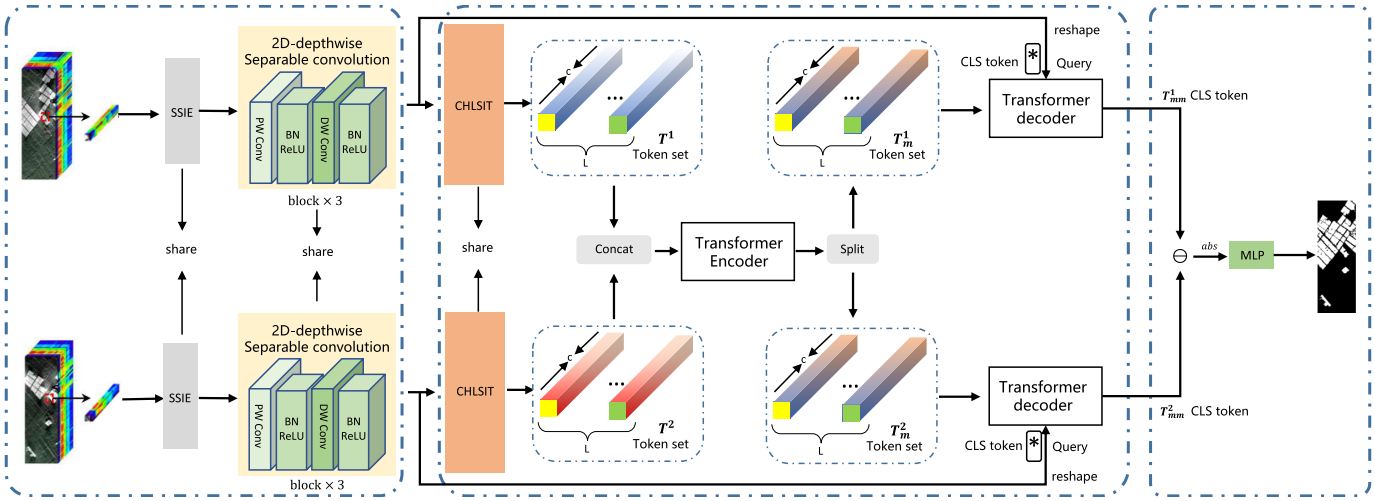


Fig. 2. Proposed framework of joint STSS-Net. We first use the SSIE module to process HSI patches to preserve key spatial features of prominent centers and similar spectra. Then, the CHLSIT module aggregates the image features extracted from CNN into a compact high-level spectral token set and provides these dual time series spectral tokens to the TE to associate information changes in the temporal and spectral sequences. The obtained dual temporal–spectral tokens are combined with pixel spatial features in the transformer decoder (TD) for temporal–spectral–spatial features. Finally, our prediction header generates pixel-level predictions by calculating the changes in CLS tokens and feeding them to MLP.

key modules: a CNN for saliency spatial feature extraction, transformer architecture for spectral redundancy removal, and a prediction head.

2) We have designed a spatial saliency information extraction (SSIE) module for denoising to improve spatial saliency. Based on distance and spectral similarity, the ability to extract global information can be improved by expanding the limited patch size and selecting the most suitable local-global information feature extraction effect.
3) We have designed a compact high-level spectral information tokenizer (CHLSIT) that removes redundancy by extracting high-level spectral conceptual features, improving the saliency of spectral information. The high-level conceptual information of changes in spectral interest can be represented by nonlinear combinations of several spectral bands, combined with neighborhood spectral information to make spectral information more compact. The spectral sequence and temporal context are modeled based on high-level spectral concept tokens.

The remaining content is presented as follows. Section II provides an elaboration on related works regarding HSI-CD. Section III outlines the framework of the STSS-Net for HSI-CD. Section IV presents three datasets and experimental results. Finally, Section V introduces the conclusion.

II. Related Works

A. HSI-CD Based on DL

With the rapid development in the field of images in recent years, deep-learning-based methods are currently the most commonly used change detection methods, especially CNN [10], which are increasingly favored by the CD industry. Before this, some methods were extensions of traditional methods. For example, traditional change vector analysis [29], [30] and slow feature analysis [31] techniques have been adapted for DL. This has led to the introduction of deep change vector analysis (DCVA) [32]. Additionally, some researchers prefer to directly map images to changes [33], [34], [35], [36], [37].

However, most current methods focus on multispectral images or other RS images. In recent years, there has been a rise in DL techniques specifically designed for HSI-CD [38], [39], [40], [41]. In addition, other methods also treat high-dimensional hyperspectral data as tensors, which have good change detection accuracy and automation. Huang et al. [42] used tensor-based DL methods for HSI-CD. Zhou and Chen [43] introduces an HSI-CD based on self-supervised tensor networks. Focusing on convolutional networks and attention mechanisms, Wang et al. [44] have proposed an end-to-end connected CNN (SiamNet) with spectral–spatial intelligent attention mechanism (SSA-SiamNet), which can highlight information channels and positions, suppress channels and positions with less information. Additionally, Yang et al. [45] have proposed a deep multiscale pyramid network based on spatial–spectral residual attention enhancement, Improved multiscale spatial–spectral feature mining capabilities, and detection performance for complex change regions. Wang et al. [15] proposed another method called GETNET, which adopts an automated approach. It first performs spectral unmixing of linear and nonlinear features, fusing richer features. Then, each pixel is represented by a feature adjacency matrix. The features of this matrix are further extracted using CNN, and an affinity matrix is constructed using CNN. Finally, a classifier is used to determine whether the corresponding pixels have been changed.

Of course, current research also focuses more on the connection between spatial and spectral information Zhang et al. [41] fully considered the data characteristics of hyperspectral 3-D cubes and extracted features of differential images from spatial and spectral directions using the designed 3D-CNN. Ou et al. [18] have proposed a CNN framework including slow–fast band selection and feature fusion grouping (SFBSFFGNET) to address the issue of spectral band redundancy.

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It can be seen that CNN has significant CD effects in HSI feature extraction methods. In HSI-CD, the parameter-sharing mechanism of CNN also greatly alleviates the problem of insufficient HSI-CD datasets. At the same time, to improve the saliency of spatial information in CNN, the attention mechanism is very important and effective, so it is necessary to further analyze from a spatial perspective in conjunction with CNN [46]. Unlike existing spatial attention-based CD methods that directly model in pixel-based space, we also consider the similarity of substance spectra and distance from center pixels, which increases the spatial difference between similar spectral substances and different spectral substances, and reduces neighborhood noise with distance attention. Therefore, feature extraction pays better attention to changes in substances and improves spatial saliency.

B. HSI-CD Based on Transformer

In recent years, to alleviate the shortcomings of global information and sequence information capture in CNN, researchers have begun to pay attention to transformers and proposed many image-processing methods based on transformers. The transformer initially designed for natural language processing (NLP) [12] has proven effective in learning long-term sequential dependencies in input data. As a result, variants of transformers including vision transformer (ViT) [47], and DEtection TRansformer (DETR) [48], have been successfully employed in visual tasks. Moreover, transformers have gained considerable attention and achieved remarkable success in computer vision tasks, further extending their application beyond NLP.

Inspired by the above work, transformer was introduced into the field of HSI, such as HSI-CD and HSI classification, HSI anomaly detection, HSI object detection, etc. For example, Xue et al. [49] further applied transformers to multimodal data collaborative classification, namely hyperspectral and LiDAR data classification, modifying the original ViT to extract spatial features of HSI. Xue et al. [49] extended the application of transformers by utilizing them for the collaborative classification of HSI. Hong et al. [25] can learn local sequence information of the spectrum from adjacent bands of HSIs. Chen et al. [28] utilized transformers to capture the remote context in a compact token-based spatial–temporal domain. Wang et al. [50] designed a converter from a multiscale perspective to learn features at different scales for CD. Wang et al. [23] proposed a spectral–spatial–temporal transformer for HSI-CD. Dong et al. [51] proposed a CD network called D-LIEG to enhance local global features.

Transformers have the advantage of modeling global sequence information, and we have decided to use a TE to extract spectral sequence features. Because the transformer method designed for HSIs requires a series of regular tokens to be received, which requires all pixels to be tiled into a vector. Therefore, directly converting them will result in redundancy of spectral information and loss of neighborhood contextual spectral information. Therefore, we propose a tokenizer to model the global spectral dependencies of all neighboring pixels and remove redundancy by learning high-level spectral information. The dual branch TE and decoder network of HSI-CD can further improve the change detection ability of the transformer by injecting spatial information, spectral information, and temporal information, thereby modeling the temporal–spectral–spatial features of HSI. In this article, we explore the advantages of transformers in HSI, while complementing the advantages of CNN to achieve better results.

III. The Proposed Method

Due to the sequence and redundancy of spectra, as well as the limitations of patch-based methods, we want to utilize the advantages of CNN and transformers to extract more salient spatial and spectral information, respectively. Inspired by this, we propose a joint STSS-Net for HSI-CD. The entire process of our STSS-Net is shown in Fig. 2.

Our STSS-Net architecture has three main components.

1) SSIE module, which enhances the saliency of spatial information.
2) CHLSIT module, which learns the relationships between spectra and generates a set of compact high-level spectral tokens that combine neighborhood information for each time input.
3) CNN and TEs and decoders extract the spatial, temporal, and spectral information of patches, respectively. Finally, combining saliency spatial features with dual temporal high-level spectral token sets in the decoder.

In this section, we provide the details of the proposed network.

A. Architecture

The inference details of our overall change detection model are shown in Algorithm 1.

1) CNN: Our model first uses the SSIE module to preserve prominent key information, to more effectively focus on spatial saliency information. At the same time, due to the insufficient labeling of HSI-CD datasets, to alleviate the problem of overfitting, we adopted a method that utilizes depthwise separable convolution and batch normalization to reduce parameters. This unit contains pointwise convolution (PW conv) and depthwise convolution (DW conv) with 3×3-size kernel. After the first two convolutional layers, there is a batch normalization layer and a ReLU activation layer. Obtain the spatial detail feature mapping of the saliency of the input image through the convolutional network in this section.

2) TE and Decoder: Then input the feature map into the CHLSIT module, and convert the feature map into compact high-level spectral information tokens. Using spectral high-level feature learning to enhance the saliency of spectral features.

Subsequently, the dual time token sets are concatenated position encoded, and fed into the first TE to generate enhanced dual time features. Encode the position after reshaping the tokens. Finally, input it into the second TE to generate enhanced spectral sequence features. Then, the dual temporal–spectral feature tokens are separated and combined with the spatial features extracted by the CNN through TDs.

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Algorithm 1 STSS-Net

Input:
$\mathcal{I} = \{(I^1, I^2)\}$ (a pair of patches)
Output:
$\mathcal{M}$ (a change prediction map)
// step1: extract spatial saliency information
while $i = 1, 2$ **do**
  ${X_{ssi}}^i$ = SSIE(I^i)
  X^i = CNN_Backbone(${X_{ssi}}^i$)
end while
// step2: compute the token sets
while $i = 1, 2$ **do**
  T^i = Tokenizer(X^i)
end while
// step3: extract temporal–spectral sequence information
T = Concat(T^1, T^2)
T_m = Transformer_Encoder(T)
// step4: joint saliency temporal–spatial–spectral information
${T_m}^i$ = Split(T_m) ($i = 1, 2$)
while $i = 1, 2$ **do**
  ${T_{mm}}^i$ = Transformer_Decoder(X^i, ${T_m}^i$)
end while
// step5: predict changes
$\mathcal{M}$ = Prediction_Head(${T_{mm}}^1$, ${T_{mm}}^2$)

The spatial features here need to be reshaped first. Then, an extra learnable classification token is added and position encoded to extract temporal–spectral–spatial joint features. Our key is to enhance the saliency of spectral and spatial features and combine them, which is beneficial for the comparative learning feedback of dual temporal features.

3) Prediction Header: The CLS tokens are fed into the prediction header for pixel-level prediction. For the output of the decoder, we only take CLS tokens as the basis for prediction. Given that the two temporal–spectral–spatial joint feature mapping tokens from the decoder output are ${T_{\mathrm{mm}}}^1$, ${T_{\mathrm{mm}}}^2 \in \mathbb{R}^c$ (c is the number of channels in the spatial feature map after passing through the CNN), the prediction head formula is as follows:

$$M = \sigma\left(g\left(\left|{T_{\mathrm{mm}}}^1 - {T_{\mathrm{mm}}}^2\right|\right)\right) \tag{1}$$

where $g(\cdot)$ represents a fully connected layer, $M \in \mathbb{R}^2$ and $\sigma(\cdot)$ represents the softmax activation function.

Finally, during the training phase, we use cross-entropy loss to optimize network parameters. The calculation formula for the loss function is as follows:

$$L = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log \hat{y}_i + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right) \tag{2}$$

where n represents the number of samples, y_i represents the true label, and $\hat{y}_i$ represents the predicted probability value calculated by the softmax function.

B. Spatial SSIE Module

Due to the lack of labeled HSI-CD data, patch-based methods are usually considered, which establish a focus mapping method from the surrounding pixels to the central pixel. This method considers neighborhood information, thereby improving the robustness of detection. However, it is rough and it is not appropriate to handle pixels at each position equally in the patch. In fact, the label of the patch is only targeted at the center pixel. The closer the pixel is to the center pixel, the better it can depict the image features around the center pixel. At the same time, we also need to consider the role of pixels in the neighborhood that have spectra similar to the central pixel. If feature extraction only focuses on distance, it will reduce the robustness of neighboring pixels. Based on the similarity of substance spectra, the spatial differences between similar spectral substances and different spectral substances can be increased, thus extracting better spatial variation features. Our spatial attention module is anticipated to have the capability to dynamically regulate the extent of the region that is most vital for identifying changes in central pixels, eliminating noise, and enhancing saliency information.

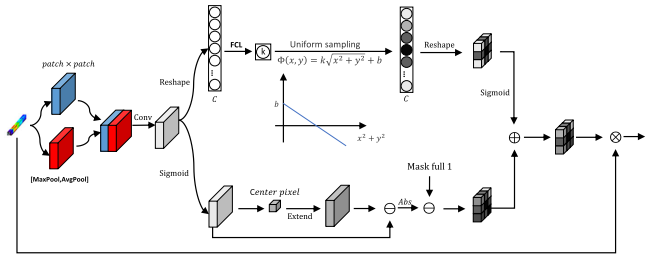


Fig. 3. Spatial SSIE module.

As shown in Fig. 3, set I^1, $I^2 \in \mathbb{R}^{C\times p\times p}$ as the input dual time series HSI patches, where p and C are the length and width of the truncated HSI patches, and the number of spectral channels in the HSIs. Set ${X_{\mathrm{ssi}}}^1$, ${X_{\mathrm{ssi}}}^2 \in \mathbb{R}^{C\times p\times p}$ as the output. To better find the differences between spectra, find the pixel positions of similar spectra in space, and prevent the search error of similar spectral substances caused by different characteristics of similar substances in a certain spectrum [52]. I^1, I^2 are first used to obtain feature maps x^1, $x^2 \in \mathbb{R}^{p\times p}$ using the maximum pooling layer, average pooling layer, and convolution layer for the spectral channel dimension. The attention map is obtained by assigning different weights to the central spectral pixel attention and similar spectral pixel attention, respectively. The attention map is then obtained by adding matrices to obtain different weights. Finally, the spatial saliency information is obtained by multiplying the original input HSI patches I^1, $I^2 \in \mathbb{R}^{C\times p\times p}$

$$x^i = \delta\left(\mathrm{Concat}\left(\mathrm{AVG}\left(I^i\right), \mathrm{MAX}\left(I^i\right)\right)\right) \tag{3}$$

$$X_{\mathrm{ssi}} = \left(W\alpha\left(x^i\right) \oplus (1 - W)\beta\left(x^i\right)\right) \otimes I^i \tag{4}$$

where $\delta(\cdot)$ represents a kernel size of 3×3. A convolutional layer with padding of 1, $\alpha(\cdot)$ represents the center pixel saliency method, $\beta(\cdot)$ represents the similar spectral pixel saliency method, $\oplus$ represents the matrix sum, $\otimes$ represents the matrix element by element multiplication, and W represents the different weights of the two methods. This article conducted experiments and found that a value of 0.5 for W is the best effect. The experiment is shown in Section IV.

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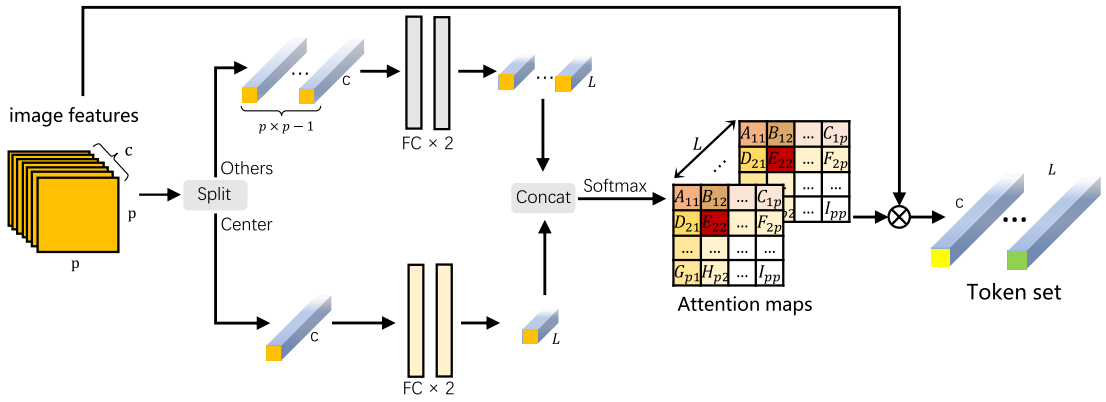


Fig. 4. Compact high-level spectral information tokenizer.

1) Preserve Center Pixel Saliency: To preserve the significant features of the central range based on distance and reduce the degree of attention from the center to the outer circumference, we plan to introduce a uniform sampling function. At the same time, to prevent the gradient from disappearing, it is necessary to measure the distance away from the square root, which is expressed in the formula as

$$\varphi(x, y) = k\sqrt{x^2 + y^2} + b \quad (k < 0) \tag{5}$$

(x, y) is the spatial coordinate of the pixel relative to the center pixel, k is the rate of descent, and b is the value of the center pixel. At this point, k controls the decreasing rate of attention, and the fully connected layer can learn the corresponding k for each patch, thereby achieving adaptive attention.

To meet the input form, it is necessary to first reshape the feature maps x^1, $x^2 \in \mathbb{R}^{p\times p}$ into 1-D vectors v^1, $v^2 \in \mathbb{R}^{pp}$. In addition, considering the non-positive nature of uniform sampling slope and the possibility of negative numbers due to monotonic decreasing. We use the Softplus activation function for activation. The above formula is expressed as

$$\hat{k} = (-1)\gamma\left(\hat{w}^T\hat{v} + \hat{b}\right) = (-1)log\left(1 + e^{\hat{w}^T\hat{v}+\hat{b}}\right) \tag{6}$$

where $\hat{w}$ and $\hat{b}$ represent the weights and biases of the fully connected layer, and $\gamma(\cdot)$ represents the SoftPlus activation function.

By sampling with $\varphi(x, y)$, the contribution of each position pixel to change detection can be obtained. For the obtained vectors v^1, v^2, reshape as x^1, $x^2 \in \mathbb{R}^{p\times p}$. In actual calculations, the attention score range should be 0-1, so use the sigmoid activation function for activation. The overall formula is expressed as

$$\text{Attention_Map}^1 = \sigma\left(\zeta\left(\mathcal{F}\left(x^1; x^2\right)\right)\right) \tag{7}$$

where $\mathcal{F}(\cdot)$ represents the fully connected layer, $\zeta(\cdot)$ represents uniform sampling, and $\sigma(\cdot)$ represents the sigmoid activation function.

2) Preserve the Saliency of Similar Spectral Pixels: Unlike the method of preserving the saliency of the central pixel, to ensure that the attention score range is within 0-1, we need to first activate the input x^1, $x^2 \in \mathbb{R}^{p\times p}$ using the sigmoid function.

Then obtain the spectral feature value of the central pixel $b \in \mathbb{R}$, expand it to $b \in \mathbb{R}^{p\times p}$, and compare it with the original feature maps x^1, $x^2 \in \mathbb{R}^{p\times p}$. Calculate the absolute value of the obtained feature map. The value range is 0–1. The larger the value, the greater the spectral difference. Finally, subtract with a mask filled with 1 to obtain an attention map. The higher the attention score, the greater the spectral similarity between the two. The above formula can be expressed as

$$\text{Attention_Map}^2 = \text{MASK} - |\sigma\left(x^1; x^2\right) - \text{Expend}(b)| \tag{8}$$

where $\sigma(\cdot)$ represents the sigmoid activation function.

C. Compact High-Level Spectral Information Tokenizer

The HSI-CD dataset has the problem of insufficient dataset labeling, so for spatial information, the feature learning of each pixel is precious. However, the spectral redundancy caused by the excessive number of spectral channels is the biggest problem that affects the efficiency of feature extraction and CD models. Before extracting spectral sequence through VIT, the feature maps extracted by convolutional networks need to be reshaped as a token set, but this method cannot remove redundancy. This module aims to remove redundancy by extracting high-level spectral features, and then combining high-level spectral concept feature information (i.e. attention map) with neighborhood high-level spectral features through matrix multiplication to obtain tight high-level spectral tokens.

As shown in Fig. 4, let X^1, $X^2 \in \mathbb{R}^{c\times p\times p}$ be the dual time series feature maps extracted through the CNN section, respectively. Let Y^1, $Y^2 \in \mathbb{R}^{L\times p\times p}$ be the obtained spectral high-level feature attention map, where L is the length of the token set and the number of spectral high-level conceptual features. Let T^1, $T^2 \in \mathbb{R}^{L\times c}(L < c)$ be the token set generated after passing through the CHLSIT module.

To learn the high-level spectral feature information between each spectrum of each pixel point and avoid interference caused by the interaction between the central pixel learning and the background part, the feature maps X^1, X^2 are first separated from the central pixel and background neighboring pixels. As shown in Fig. 4, the $p \times p - 1$ pixel spectral sets are first used to learn the relationship between the spectra of each pixel point in the background through two fully connected layers. This allows for better nonlinear learning of

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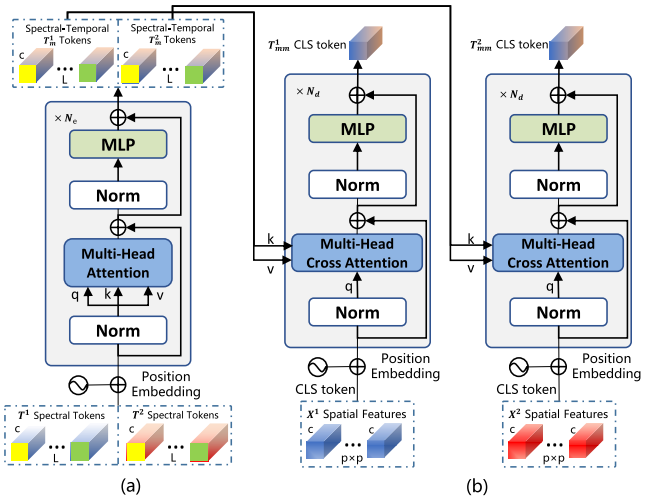


Fig. 5. Transformer architecture. (a) Transformer Encoder. (b) Transformer Decoder.

high-level spectral features in neighboring backgrounds. similar to channel attention, but our channel count is reduced from c to L. Similarly, for the central pixel, obtain the high-level spectral features of the central pixel. For the high-level spectral features, concatenation and softmax activation functions are used to calculate the spectral high-level feature channel attention map $Y^i (i = 1, 2)$ in the spectral channel dimension.

As shown in Fig. 4, if the attention icon is identified as the same letter, it represents the spectral high-level concept features after spectral correlation learning under this pixel point, and L represents the number of nonlinear bands represented by the high-level features after spectral compression. Finally, an attention graph is used to multiply the matrix with X^1, $X^2 \in \mathbb{R}^{c\times p\times p}$, so that each point in the token set contains neighborhood high-level feature spectral information. Finally, we obtain high-level spectral tokens containing neighborhood-tight information that is de redundant. The overall formula can be expressed as follows:

$$T^i = Y^i \otimes \left(X^i\right)^T = \sigma\left(g\left(X^i\right)\right) \otimes \left(X^i\right)^T \tag{9}$$

where $g(\cdot)$ represents two fully connected layers, $\sigma(\cdot)$ represents the softmax activation function, and $\otimes$ represents matrix multiplication.

D. Spectral–Spatial–Temporal Feature Union in Transformers

1) TE (Spectral–Temporal Features): Generally speaking, a converter is a structure that has three main modules: position embedding, self-attention (SA), and multilayer perceptron (MLP). The SA mechanism captures the internal correlations of temporal and spectral sequences by embedding the order of captured input sequences. This article captures global time series and spectral sequence information through the TEs.

As shown in Fig. 5(a), after obtaining the two spectral token sets T^1, $T^2 \in \mathbb{R}^{L\times c}$ of the input dual time series image, we first concatenate T^1, T^2 to obtain $T \in \mathbb{R}^{2L\times c}$. After positional encoding, we use a TE to model the context of the dual time series, and then reshape to obtain the token set $T \in \mathbb{R}^{c\times 2L}$. After position encoding, we use a TE to model the context of these spectral sequences. Obtain a new token set ${T_m}^1$, ${T_m}^2 \in \mathbb{R}^{L\times c}$, which includes joint spectral–temporal features for comparative learning.

The layer normalization occurs before MSA/MLP. $T^0 \in \mathbb{R}^{2L\times c}$ is input into the TE composed of MSA and MLP alternately to encode the time series. Same modeling process as for spectral sequences. At each layer l, the formula can be expressed as

$$\begin{aligned} \left(T^l\right)' &= \mathrm{MSA}\left(\mathrm{LN}\left(T^{l-1}\right)\right) + T^{l-1} \\ T^l &= \mathrm{MLP}\left(\mathrm{LN}\left(\left(T^l\right)'\right)\right) + \left(T^l\right)' \end{aligned} \tag{10}$$

where LN is the normalization layer.

The input of the MSA is a triplet (query Q, key K, value V) calculated from the input $T^{l-1} \in \mathbb{R}^{2L\times c}$, where the MSA formula can be expressed as

$$\begin{aligned} Q &= T^{l-1} W^q \\ K &= T^{l-1} W^k \\ V &= T^{l-1} W^v \end{aligned} \tag{11}$$

$$\mathrm{Att}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{12}$$

$$\begin{aligned} \mathrm{MSA}(T) &= \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h) W^O \\ \mathrm{head}_j &= \mathrm{Att}\left(T W_j{}^q, T W_j{}^k, T W_j{}^v\right). \end{aligned} \tag{13}$$

W^q, W^k, $W^v \in \mathbb{R}^{c\times d}$, $W^O \in \mathbb{R}^{hd\times c}$ are the learnable parameters of the three linear projection layers, h is the number of SA heads, and d represents the channel dimension of the triplet.

2) TD (Temporal–Spectral–Spatial Features): It is crucial to combine spatial and spectral features that are more significant after redundancy removal for HSI-CD. So far, as shown in Fig. 5(b), we have obtained two sets of contextually rich temporal–spectral token sets ${T_m}^1$, ${T_m}^2 \in \mathbb{R}^{L\times c}$ for each time image. Now, we need to combine the token set with pixel spatial features to obtain temporal–spectral–spatial features. To achieve this, we removed the MSA block and replaced it with cross-attention. In MSA, the queries, keys, and values come from the same input sequence. In cross attention, the Q in the triplet comes from the spatial features X^1, $X^2 \in \mathbb{R}^{c\times p\times p}$, and K and V come from the spectral feature token set ${T_m}^1$, ${T_m}^2$. Then, the spatial feature map is reshaped and position encoded, and a CLS token is added. The formulas for the differences can be expressed as

$$\begin{aligned} Q &= X^{l-1} W^q \\ K &= {T_m}^{l-1} W^k \\ V &= {T_m}^{l-1} W^v \end{aligned} \tag{14}$$

$$\begin{aligned} \mathrm{MCA}(T_m, X) &= \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h) W^O \\ \mathrm{head}_j &= \mathrm{Att}\left(X W_j{}^q, T_m W_j{}^k, T_m W_j{}^v\right). \end{aligned} \tag{15}$$

IV. Experiments

We analyze the impact of STSS-Net through experiments in this section. First, we describe three datasets. Second, the experimental setup was introduced, including STSS-Net hyperparameters, experimental environment, and evaluation criteria. Third, we compared the performance of the proposed method with other state-of-the-art HSI-CD algorithms. Fourth,

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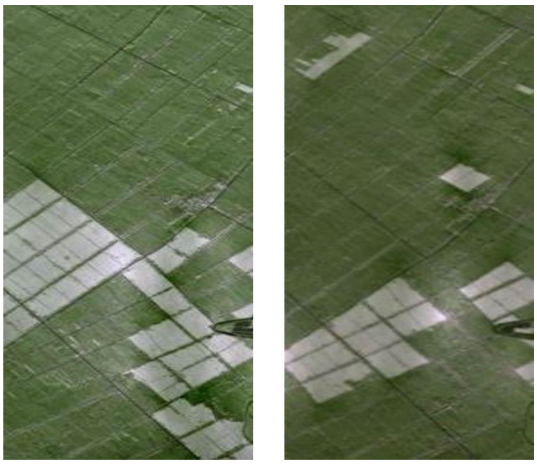


Fig. 6. Farmland Dataset.

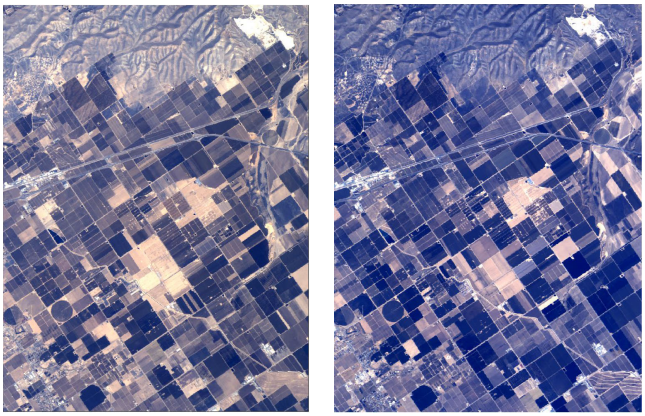


Fig. 7. Santa Barbara Dataset.

we reveal the innovation of SSIE and CHLSIT through ablation experimental analysis. Fifth, the parameters of STSS-Net were analyzed.

A. Experimental CD Datasets

1) Farmland Dataset: As shown in Fig. 6, the dual temporal images of the dataset were captured in Jiangsu Province, China in 2006 and 2007, respectively. After removing noise and water absorption bands, HSI contains 155 bands. Its size is 420 × 140 pixels. It mainly includes changes in farmland and detection of changes in farmland.

2) Santa Barbara Dataset: The dataset used in this study, shown in Fig. 7, consists of bitemporal HSIs collected in 2013 and 2014 over the Santa Barbara region in California using the AVIRIS sensor. Each image has dimensions of 984 × 740 pixels and contains 224 spectral bands. Importantly, this dataset includes 52 134 pixels that have changed, 80 418 pixels that have remained unchanged, and 595 608 pixels with unknown status.

3) Bay Area Dataset: Acquired in 2013 and 2014 using the AVIRIS sensor, the Bay Area dataset encompasses bitemporal HSIs. This dataset specifically covers the Patterson area in California. With a 20 m spatial resolution, the dataset has dimensions of 600 × 500 pixels. It captures information across 224 spectral bands in the range of 0.4–2.5 μm. Notably, the dataset includes 38 425 changed pixels, 34 211 unchanged pixels, and 227 364 pixels with unknown status. Fig. 8 illustrates the RGB images obtained from the Bay Area dataset.



Fig. 8. Bay Area Dataset.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THREE DATASETS

Dataset	Unchange		Change	
	Training	Test	Training	Test
Farmland	500	39917	500	17883
Santa Barbara	79918	500	45.66	51634
Bay Area	500	33711	500	37925

To obtain better results in limited samples and better compare with other models, we divided the number of training pixels and test pixels as shown in the Table I.

B. Experimental Setup

1) Implementation Details: We implemented our proposed method using Intel® Xeon[1] Silver 4210R CPU@2.40 GHz × 40 CPU and NVIDIA GeForce RTX 3090 GPU on PyTorch and MindSpore. We use stochastic gradient descent (SGD) with a momentum of 0.90 and weight decay of 0.0005 to optimize the model. The initial learning rate is set to 0.0005. Meanwhile, the learning rate is multiplied by 0.9 for every 1/20 total epoch. There are a total of 200 epochs. 9 is the size of the patch, and 16 is the size of the batch. The L of the spectral high-level concept tokens is 8, where the number of temporal encoders is 1, the number of spectral sequence encoders is 1, the encoders are in series, and the number of decoders is 4, all of which have positional encoding.

2) Evaluation Metrics: Using overall accuracy (OA) and kappa coefficient (KC) as evaluation indicators, the calculation formula is as follows:

$$\mathrm{OA} = \frac{\mathrm{TP+TN}}{\mathrm{TP+FP+TN+FN}} \tag{16}$$

$$\mathrm{Pc} = \frac{(\mathrm{TP+FP})(\mathrm{TP+FN})(\mathrm{TN+FN})(\mathrm{TN+FP})}{(\mathrm{TP+FP+TN+FN})^2} \tag{17}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA-Pc}}{1-\mathrm{Pc}} \tag{18}$$

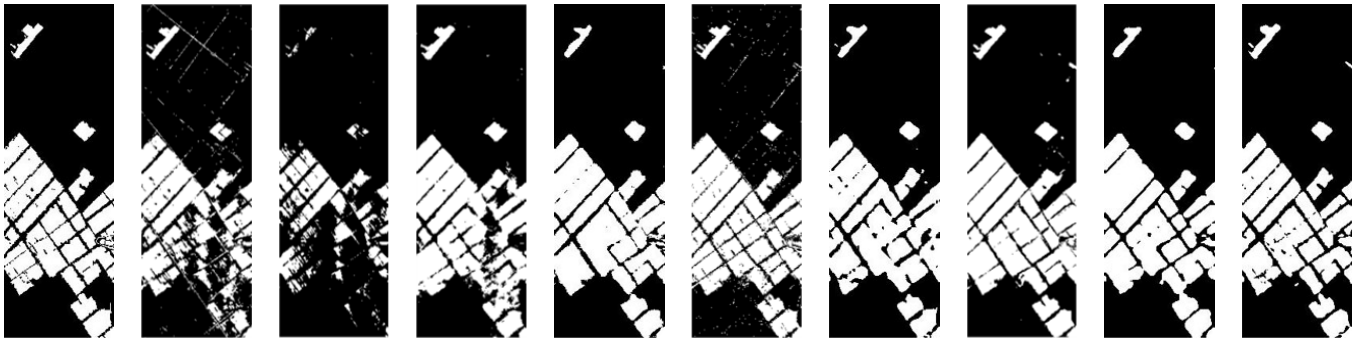
where true positive (TP) in this article represents the changed pixels that are correctly recognized in the classifier's prediction results, true negative (TN) in this article

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TABLE II
COMPARISON OF TEST RESULTS OF EXCELLENT HSI-CD METHOD ON THREE DATASETS

Method	Farmland		Santa Barbara		Bay Area	
	OA(%)	KC	OA(%)	KC	OA(%)	KC
CVA	87.49	69.98	83.25	65.19	85.17	70.63
DPCA	81.00	48.09	75.66	44.03	67.77	37.90
SVM	90.87	78.97	97.01	93.77	90.51	80.94
CNN	94.48	88.63	95.35	90.12	95.81	91.59
GETNET	94.03	86.40	97.37	94.48	97.49	94.72
BIT-CD	95.14	88.75	98.50	96.85	97.10	94.19
SST-Former	95.21	88.43	98.10	96.58	97.89	95.94
Base	94.89	87.91	97.55	95.82	97.03	95.21
Proposed	**96.20**	**91.16**	**98.56**	**96.97**	**98.30**	**96.58**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. CD results on the Farmland dataset. (a) Ground truth. (b) CVA. (c) DPCA. (d) SVM. (e) CNN. (f) GETNET. (g) BIT-CD. (h) SST-Former. (i) Base. (j) STSS-Net. The white area is changed, and the black area is unchanged.

represents the unchanged pixels that are correctly recognized in the classifier's prediction results, false positive (FP) in this article represents the changed pixels that are falsely reported in the classifier's prediction results, and false negative (FN) in this article represents unchanged pixels that have been falsely reported in the classifier's prediction results.

C. Comparison to the State-of-the-Art

This article compares six excellent algorithms, namely CVA [30], DPCA [53], SVM, CNN [10], GETNET [15], BIT-CD [28], and SST-Former [23], to verify the effectiveness of our algorithm. The patch size for CVA, DPCA, and BIT-CD is 5, while the patch size for SST-Former is 9. The other details of these algorithms are as follows.

CVA obtains change vectors and automatically classifies changes and nonchanges using K-means.

DPCA named differential principal component analysis obtains the difference map of dual time HSI and detects changes through K-means.

SVM uses principal component analysis to extract 10 principal components from the difference map and trains the support vector machine with 500 changing pixels and 500 invariant pixels, respectively.

CNN, a CD method combining CNN, sparse autoencoder, and clustering.

GETNET uses linear and nonlinear methods for hyperspectral unmixing and further extracts the features of the matrix for classification through CNN. We train GETNET with other parameters being the same as the original text.

BIT-CD uses CNN and transformers as backbone networks. We retain the architecture and parameters of the original model. To adapt to HSIs, we modify the residual network to the same CNN as in this article (deep separable convolution and point convolution).

SST-Former has designed a new transformer architecture that learns spectral information in spatial, spectral, and temporal domains for change detection and classification. The parameters are consistent with the original text.

Base, our baseline consists of a CNN backbone, a VIT, and a prediction header, replacing the cross attention in the decoder with SA. Remove SSIE and replace the CHLSIT module with an adaptive average pooling layer, then reshape to obtain the token set, keeping the shape of the input data of the TE unchanged.

To verify the effectiveness of our proposed method, we visually and quantitatively compared it with other mature methods. Table II reported the data comparison results of three datasets. Figs. 9–11 show the visual comparison results of three datasets, respectively.

The quantitative results indicate that the model based on CNN and transformer performs better than other methods on these datasets. For example, our STSS-Net kappa scores

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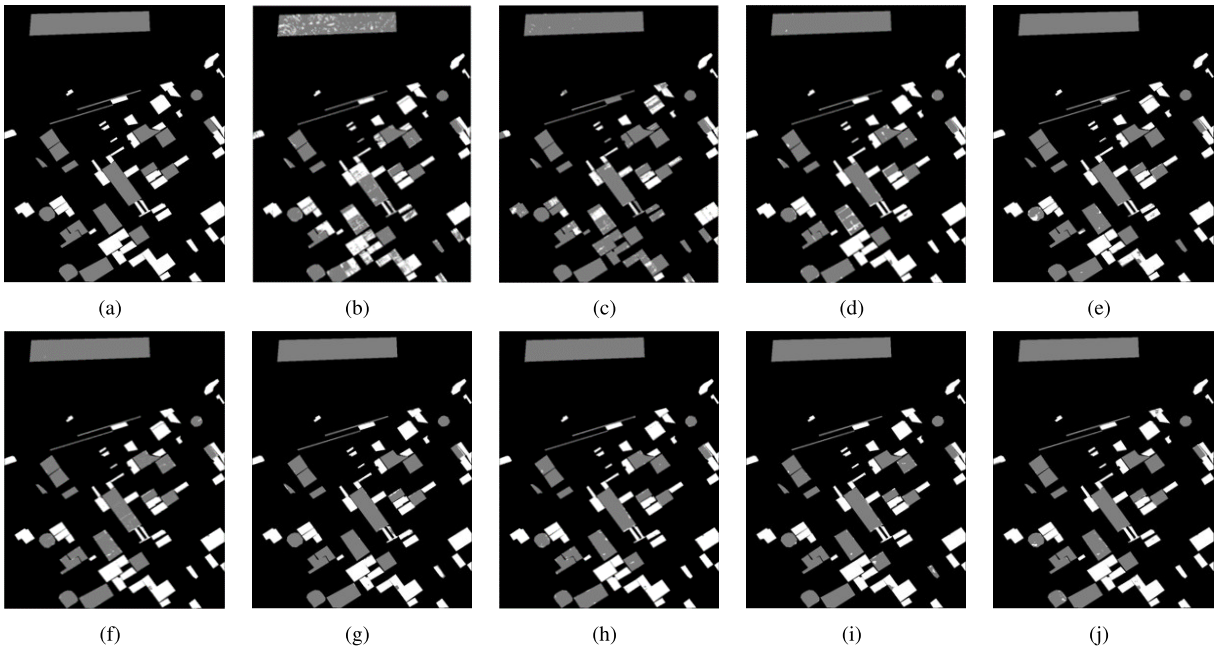


Fig. 10. CD results on the Santa Barbara dataset. (a) Ground truth. (b) CVA. (c) DPCA. (d) SVM. (e) CNN. (f) GETNET. (g) BIT-CD. (h) SST-Former. (i) Base. (j) STSS-Net. The white is the changed area. The gray is the unchanged area. Black is the unknown area.

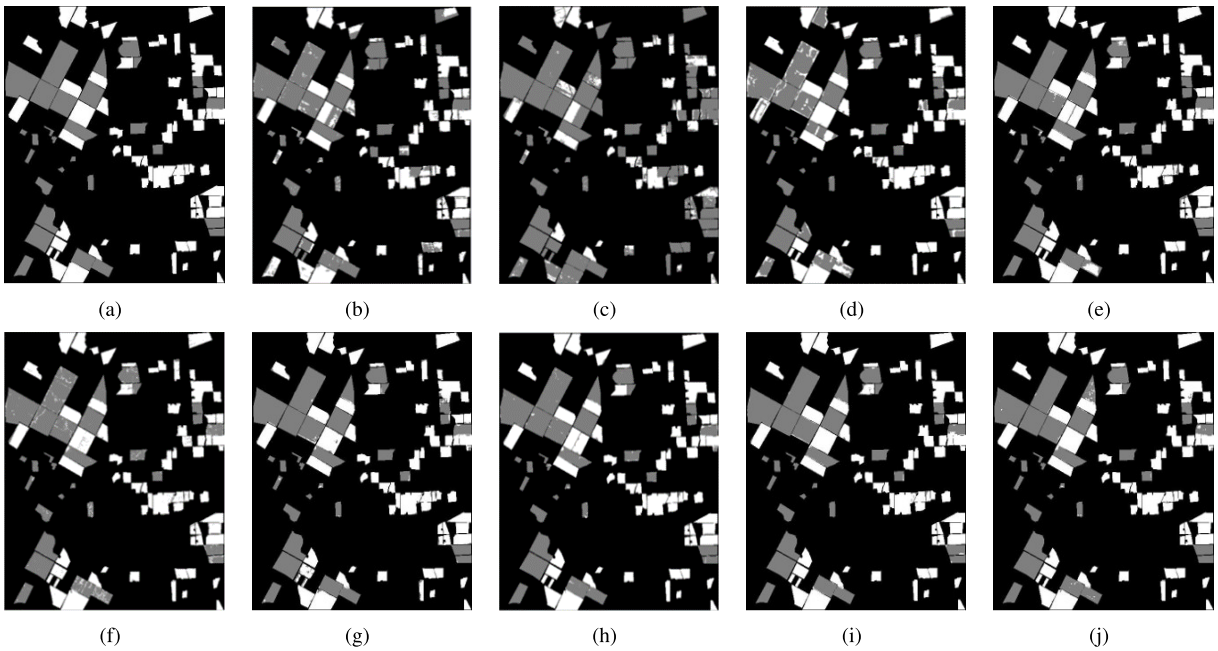


Fig. 11. CD results on the Bay Area dataset. (a) Ground truth. (b) CVA. (c) DPCA. (d) SVM. (e) CNN. (f) GETNET. (g) BIT-CD. (h) SST-Former. (i) Base. (j) STSS-Net. The white is the changed area. The gray is the unchanged area. Black is the unknown area.

were 2.73/0.39/0.64 higher than the latest SST-Former on three datasets, respectively. The OA scores exceeded the latest SST-Former scores of 0.99/0.46/0.41, respectively. Meanwhile, our STSS-Net kappa scores improved by 3.25/1.15/1.37 compared to baseline, respectively. The OA scores exceeded the baseline scores of 1.31/1.01/1.27, respectively. This reflects the effectiveness of our SST-Net model in HSI-CD tasks.

At the same time, some methods (such as CNN, and GETNET) are based on clustering or spectral unmixing methods to obtain more refined change detection feature maps through CNN. The experimental results show that the performance of these methods is superior to traditional mathematical methods (e.g., CVA, DPCA, SVM). Our proposed method improves the saliency of spatial information through the SSIE module and extracts spatial features through CNN. Compared with CNN-based methods, STSS-Net more effectively utilizes neighborhood information and removes spatial noise. At the same time, some methods based on transformer and CNN (e.g., BIT-CD, SST- Former, and Base) obtain spectral and spatial information by utilizing the advantages of modeling local features and long sequence information using both networks, respectively. Experimental results show

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that these methods outperform methods that only use CNN. STSS-Net not only extracts spatial and spectral information through these two networks but also combines them with time series information to further improve the performance of change detection.

Among them, we found that STSS-Net significantly improved the KC in the Farmland dataset, possibly because the other two datasets had unknown regions and the contextual label information in the neighborhood of the Farmland dataset was more sufficient [54]. Our method is capable of contextual modeling of spectral–spatial–temporal significance information, making more effective use of neighboring spectral and spatial information. SSIE and CHLSIT effectively utilize neighborhood spectral spatial information, enabling the full utilization of farmland datasets with richer neighborhood information.

The visual comparison effect is shown in the Figs. 9–11. We can see that STSS-Net achieved better visual effects than other models. CVA has many incorrectly identified areas of change and unchanged areas. After the introduction of DL in DPCA, the visualization effect has been significantly improved. SVM has made some improvements, but there are still many areas of recognition errors. DL methods such as CNN and GETNET have significantly improved visual effects. After introducing transformers, BIT-CD and SST Former have higher kappa than other DL-based algorithms. Compared with other DL algorithms, the STSS-Net algorithm performs better in detecting complex situations where unchanged and changing regions are mixed. As shown in Figs. 9–11(i) and (j), after adding SSIE and CHLSIT, the local information extraction effect is improved, and the change detection effect of edges and complex pixels is improved.

The DL model can automatically learn the most informative feature representations from the original data, without the need for manual feature extraction methods. This effectively extracts information features from HSI and improves the accuracy of CD. In summary, compared with supervised algorithms, CVA still has a significant performance gap in HSI-CD tasks. SVM has shown significant improvement, but it is still not as good as DL-based methods. The DL model can automatically learn the most informative feature representations from the original data, without the need for manual feature extraction methods. This effectively extracts information features from HSI and improves the accuracy of CD. Using transformers to combine saliency information, STSS-Net has higher accuracy and kappa than other methods and generates grayscale images with fewer noise points.

D. Ablation Studies

1) Spectral–Spatial–Temporal Context Modeling: We performed ablation on the TE and TD to verify their effectiveness in temporal–spectral–spatial context modeling. From Table III, When temporary TE (T-TE) for temporal context modeling was removed from STSS-Net, we observed a consistent decrease in kappa scores for the three datasets. This indicates the importance of the SA mechanism in TE models for temporal relationships based on high-level spectral token sets.

TABLE III

ABLATION STUDY OF OUR TE AND DECODER ON THREE HSI-CD DATASETS. THE DATA IS KC. THE DEPTH OF T-TE AND S-TE ARE SET TO 1. THE DEPTH OF TD ARE SET TO 4

T-TE	S-TE	TD	Farmland	Santa Barbara	Bay Area
✓	✓	✓	91.16	96.97	96.58
×	✓	✓	90.11	96.70	96.33
✓	×	✓	89.64	96.65	96.21
✓	✓	×	87.96	93.67	92.30
×	×	✓	87.42	94.27	93.85
×	×	×	85.48	91.13	91.40

TABLE IV

ABLATION STUDY OF OUR SSIE MODULE AND CHLSIT MODULE ON THREE HSI-CD DATASETS. THE DATA IS KC

SSIE	CHLSIT	Farmland	Santa Barbara	Bay Area
✓	✓	91.16	96.97	96.58
×	✓	90.55	96.60	96.31
✓	×	89.83	95.91	95.31
×	×	89.61	95.79	95.16

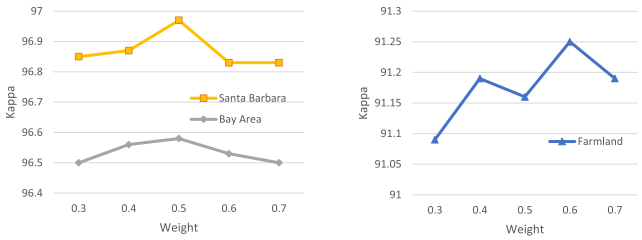


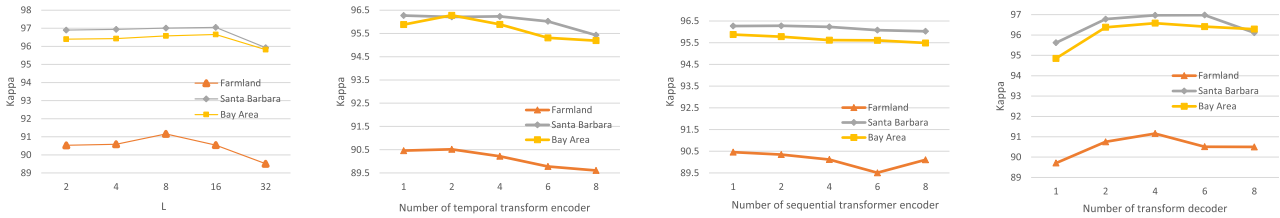
Fig. 12. Kappa scores of STSS-Net under different weight of SSIE in three datasets.

When removing sequential TE (S-TE) for spectral sequence context modeling from STSS-Net, we can observe that the KCs of the three datasets decreases more compared to removing TE for temporal context modeling, indicating the importance of modeling spectral sequence.

For the ablation of the TD, we replace TD with a simple module to fuse spectral–temporal tokens from TE and feature maps from the CNN backbone network. We extend the spatial dimension of each token set in $T_m \in \mathbb{R}^{L\times c}$ to the shape of $\mathbb{R}^{pp}$. Then tokens are multiplied by the original spatial feature map and fed into the prediction header, Table III indicates that the performance of the model without TD has consistently decreased significantly across the three test sets. This may be because cross-attention provides a way to combine spatial information, resulting in a higher information density, making it easier to effectively extract changes of interest.

2) Ablation on SSIE and CHLSIT: We ablated the CHLSIT module by removing it from the model. The conversion of tokens is achieved using an adaptive average pooling layer, and then reshaping the spatial dimension to $\mathbb{R}^L$ as the number of tokens. For example, As shown in Table IV. There is a consistent decrease in kappa scores. It indicates that the CHLSIT module is crucial for improving spectral saliency and paying attention to neighborhood spectral information in our framework based on CNN and transform. This may be because dense spectral features contain too much redundant

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(a) (b) (c) (d)

Fig. 13. Kappa scores of STSS-Net under different parameters in three datasets. (a) Token length. (b) Number of temporal TE. (c) Number of Sequential TE. (d) Number of TD.

information, making the training of transformer-based models a challenging task. By removing the SSIE module and performing ablation on it, there was a consistent decrease in kappa scores, indicating its effectiveness in improving spatial saliency.

At the same time, we eliminated the CHLSIT module and SSIE modules to explore the mutual influence between them. We found that in the process of starting from scratch, the detection effect improved by adding two modules at the same time is greater than that of adding a single module. From the perspective of unifying neighborhood information, we focus on the key information and neighborhood information of the input image patch. Specifically, the SSIE module adaptively learns and selectively focuses on important information in the input image. At the same time, CHLSIT needs to utilize advanced spectral information integrated into the neighborhood to introduce the spectral vector space, and the SSIE module can effectively help the CHLSIT module suppress the noise impact of background spectral information. In the process of representing HSI features, our proposed module uses SSIE to purify the feature space and CHLSIT to optimize the spectral feature space. Therefore, adding two modules simultaneously can greatly improve the accuracy of change detection.

E. Effect of Parameters

1) Weight of SSIE: The SSIE module has two criteria for extracting spatial saliency information, one is pixel distance, and the other is spectral similarity. To find the optimal weight for the sum of the two methods, we tested the weights = {0.3, 0.4, 0.5, 0.6, 0.7} for the significance of five different sizes of distance center pixels. As shown in Fig. 12, we found that the best effect is achieved when the weight is 0.5.

2) Token Length: Our CHLSIT module spatially aggregates the high-level features of the spectrum into a compact token set. The changes of interest in dual temporal images can be described by several high-level spectral concepts, and the band length of the high-level spectrum, i.e. the number of Token sets L. It is a crucial hyperparameter. We tested different $L = \{2,4,8,16,32\}$ to analyze their impact on model performance on the Farmland, Santa Barbara, and Bay Area datasets. As shown in Fig. 13(a), when L is quantity 8, the CD accuracy is highest in the Farmland dataset. When L is quantity 16, the CD accuracy is highest in the Farmland dataset. This indicates that a compact set of spectral tokens is sufficient to represent the high-level concepts of changes of interest, and spectral redundancy may affect the performance of the model. When L is too short, the model may lose some useful information related to changing concepts. The reason for the different effects of the two datasets at different lengths may be that the Farmland spectral channel number is 155, while the other two datasets have a spectral channel number of 224, resulting in a different number of high-level spectral concept representations. Because the difference between the two datasets is small when L is 8 and 16, L is set to 8 in this article.

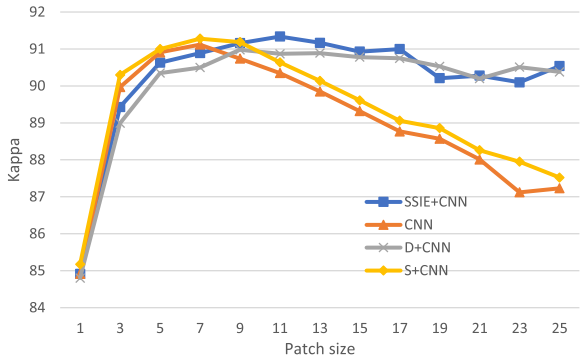


Fig. 14. Kappa curves with respect to patch size.

3) Depth of Transformer: The number of transformer layers is an important hyperparameter. To find the optimal number of two types of encoders and decoders, we tested eight different numbers $n = \{1, 2, 4, 6, 8\}$ to find the optimal number of encoders and decoders. As shown in Fig. 13(b)–(d), we found that different datasets have different optimal levels. In this article, we choose 1 as the temporary TE and 1 as the final number of spectral sequence TE. Number of 4 decoders.

F. Interpretability Analysis

1) SSIE: As shown in Fig. 14, this curve represents the changes in kappa under different patch sizes in the Santa Barbara dataset. To better explain the spatial effects of the SSIE module, we removed the transformers in the Base to prevent interference from spectral information. Only the spatial feature extraction module of CNN is retained, and finally, the obtained feature map is reshaped into a 1-D vector input to the prediction head. It can be intuitively seen that as the patch size increases, the results of CNN will become rougher and rougher. This is mainly because the proportion of neighborhood information is too large, resulting in increased noise, reduced attention to the center pixel, and reduced differences

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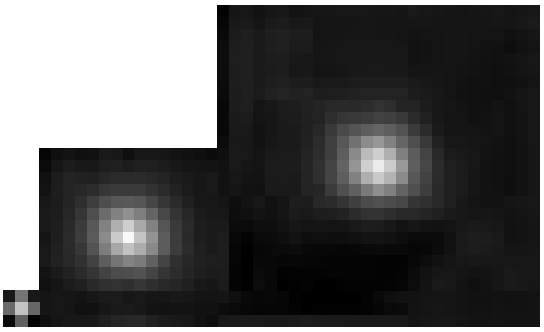


Fig. 15. SSIE attention score heat maps corresponding to three representative patches (3, 15, 27), where k is the standard deviation (-0.6903, -0.6613, -0.6284).

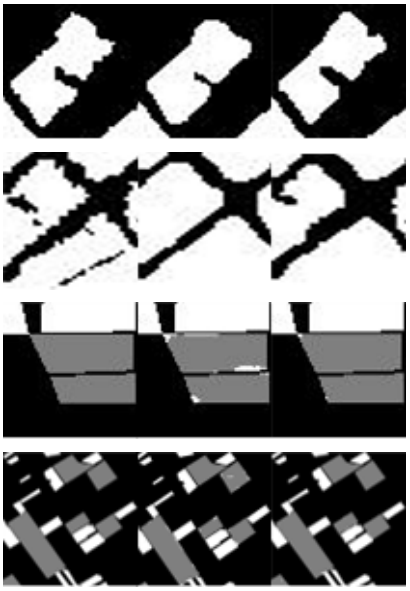


Fig. 16. Display the effectiveness of predicting edge detail information (Ground truth, Ablation on CHLSIT, CHLSIT from left to right).

between different patches. For the method of preserving center pixel significance (D+CNN) in SSIE, compared to CNN, it can maintain similar granularity under different patch sizes. When the maximum kappa score is stable, the corresponding patch size is larger, which may be due to the expansion of local information; Compared to CNN, the method of preserving the significance of similar spectral pixels (S+CNN) in SSIE effectively improves accuracy.

Fig. 15 shows the normalized spatial attention score heatmap corresponding to three representative patches in the SSIE module, where k is the uniformly sampled learning coefficient. This fully demonstrates the role of the proposed attention mechanism in improving performance and stability. To facilitate the comparison of experimental and training speeds, this article only uses patch = 9.

2) CHLSIT: In the CHLSIT module, de redundancy of spectral information, interaction with neighboring spectral information, and enhancement of spectral saliency have significant effects on improving edge information and change detection in complex situations. In the prediction graph of Fig. 16, the effect of edge information extraction can be intuitively seen for the ablation of the CHLSIT module in the three datasets.

V. Conclusion

In this article, we propose a CD network called STSS-Net. In STSS-Net, a spatial attention module SSIE is proposed for denoising spatial information, which improves spatial saliency based on distance and spectral similarity. The CHLSIT module was designed for extracting saliency spectral information, which removes redundancy by extracting high-level spectral conceptual features. Simultaneously utilizing the advantages of CNN and transformer architectures to combine temporal–spatial–spectral saliency information to improve the effectiveness of CD. We used a small training dataset to verify that STSS-Net achieved the state-of-the-art performance on three datasets. And verified the improvement of detection ability for edge and complex information, as well as the stability of SSIE in patch-based methods, as well as the ability to expand local information and improve accuracy. However, the performance of our proposed network largely depends on the optimal hyperparameters. Hyperparameters corresponding to different datasets are usually different, which is not conducive to the expansion of the model to other HSI-CD datasets obtained by different sensors. Our future work will focus on the following aspects. Firstly, we will effectively combine the two types of attention information through adaptive learning methods. Secondly, we will decouple spectral information through more effective and simple methods, intuitively representing the same or different high-level spectral concept information [55]. Thirdly, we will generate spectral information through some unsupervised methods and transfer learning methods, and reduce the impact of insufficient datasets [22].

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